WORKED SOLUTION

STEP I 1999, Question 4

Pure • Regions with Moduli

STEP I 1999, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q4

The key idea

Modulus signs split the plane into regions where each bracket has a fixed sign, and in each region the condition becomes an ordinary linear (or, in the last part, hyperbolic) inequality. The other big time-saver is symmetry and translation: $|x|$ and $|y|$ make a set symmetric in both axes, so you only need to work in one quadrant and reflect; and replacing x by $x-1$ simply slides the picture one unit to the right.

Solution

Part (i)

The set is unchanged if we replace x by $-x$ or y by $-y$, so it is symmetric in both coordinate axes. In the first quadrant ($x \geq 0$, $y \geq 0$) the condition is just $x+y \leq 1$: the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$. Reflecting in both axes gives the four triangles that make up a square standing on one corner, with vertices

$$(1,0), \quad (0,1), \quad (-1,0), \quad (0,-1).$$

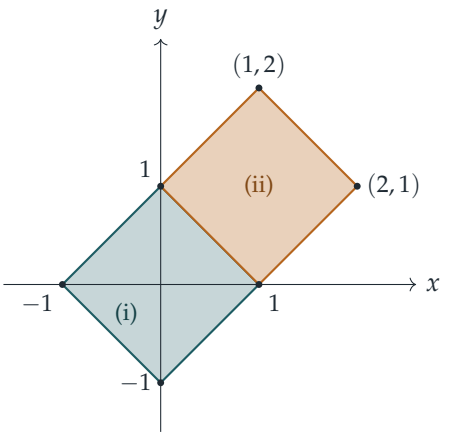
Equivalently, it is the intersection of the four half-planes $\pm x \pm y \leq 1$. The boundary is included (the inequality is $\leq$), and the area is 2.

Part (ii)

Put $X = x-1$, $Y = y-1$. Then the condition is $|X|+|Y| \leq 1$, which is part (i) in the (X,Y) coordinates. So the set is the square of part (i) translated by $(1,1)$: centre $(1,1)$, vertices

$$(2,1), \quad (1,2), \quad (0,1), \quad (1,0),$$

bounded by the lines $x+y=1$, $x+y=3$, $y=x+1$ and $y=x-1$. It touches both axes, at $(1,0)$ and $(0,1)$.



Part (iii)

Again shift: put $X = x - 1$, $Y = y + 1$, so the condition becomes

$$|X| \leq 1 + |Y|.$$

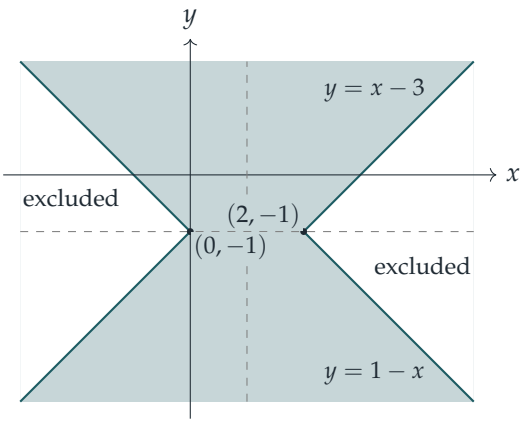
This is symmetric in both the X- and Y-axes, i.e. about the lines $x = 1$ and $y = -1$. In the quadrant $X \geq 0$, $Y \geq 0$ it reads $X \leq 1 + Y$: everything to the left of the line $X = 1 + Y$, which passes through $(1, 0)$ with gradient 1. Reflecting, the set consists of every point *except* two open wedges:

- on the right, $X > 1 + |Y|$: a right-angled wedge with apex $(X, Y) = (1, 0)$, opening to the right, bounded by $Y = \pm(X - 1)$;
- on the left, $X < -1 - |Y|$: its mirror image, with apex $(-1, 0)$, opening to the left.

Ask yourself Is this set bounded, as in (i) and (ii)?

No. The minus sign changes everything: making $|Y|$ large makes the condition *easier* to satisfy, so for each fixed Y the allowed X form the interval $-1 - |Y| \leq X \leq 1 + |Y|$, which widens as $|Y|$ grows. In particular the whole vertical line $X = 0$ (that is, $x = 1$) lies in the set.

Back in x, y coordinates the apexes are $(2, -1)$ and $(0, -1)$. The excluded right wedge is $x > 2 + |y + 1|$, bounded by the half-lines $y = x - 3$ and $y = 1 - x$ for $x \geq 2$; the excluded left wedge is $x < -|y + 1|$, bounded by $y = x - 1$ and $y = -1 - x$ for $x \leq 0$. The boundary lines themselves belong to the set. On the line $y = -1$ the set is just the segment $0 \leq x \leq 2$, its narrowest point.



The dashed lines $x = 1$ and $y = -1$ are the lines of symmetry.

Part (iv)

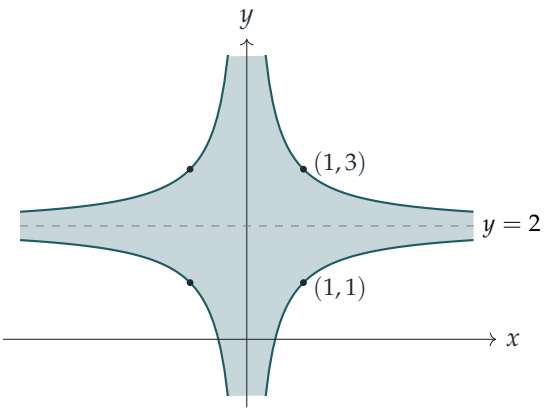
Put $Y = y - 2$, so the condition is $|x|\,|Y| \leq 1$, which is symmetric in both the x-axis and the line $Y = 0$ (that is, $y = 2$). In the quadrant $x \geq 0$, $Y \geq 0$:

- if $x = 0$ the condition holds for every Y, so the whole y-axis is in the set;
- if $x > 0$ we may divide by x, and the condition is $0 \leq Y \leq 1/x$: the region under the hyperbola $Y = 1/x$.

Reflecting in both lines of symmetry, the set is

$$x = 0, \quad \text{or} \quad 2 - \frac{1}{|x|} \leq y \leq 2 + \frac{1}{|x|} \quad (x \neq 0),$$

the region trapped between the four branches of the hyperbolas $(y-2)x = \pm 1$. It contains the whole y-axis and the whole line $y = 2$, and these are the asymptotes of the boundary curves. It looks like a fat plus sign centred at $(0, 2)$: the corners of the "square" $|x| \leq 1$, $|y - 2| \leq 1$ lie on the boundary at $(\pm 1, 1)$ and $(\pm 1, 3)$, the arms narrow towards $y = 2$ as $|x| \to \infty$, and the vertical arms narrow towards the y-axis as $|y| \to \infty$. The set is unbounded (but, curiously, has infinite area).



Checks

- Test a few points against the original inequalities. In (iii), $(1, 100)$ gives $0 - 101 \leq 1$, so it is in the set (confirming it is unbounded), while $(2.01, -1)$ gives $1.01 > 1$, just outside the right-hand apex. In (iv), $(2, 2.5)$ gives $2 \times 0.5 = 1$, on the boundary, and $(2, 2.51)$ is just outside.
- In (i), each of the four triangles has area $\frac{1}{2}$, so the square has area 2; a grid count gives 2.002.
- We tested 400,000 random points in $[-6, 6]^2$ for each part, comparing the original inequality with our description (half-planes, excluded wedges, or $|y - 2| \leq 1/|x|$); there were no disagreements.

Common slips

- In (ii) and (iii), shifting the wrong way: $|x - 1|$ moves the picture to $x = +1$, and $|y + 1|$ moves it to $y = -1$.
- In (iii), drawing a bounded diamond by analogy with (i). The minus sign reverses the role of $|y + 1|$, and the set is unbounded.
- In (iv), dividing by $|x|$ without treating $x = 0$ separately, and so missing the fact that the whole y-axis belongs to the set.

Beyond STEP

The set in (i) is the unit ball for the "taxicab" norm $\|(x, y)\|_1 = |x| + |y|$, and (ii) is the ball of radius 1 centred at $(1, 1)$. Different norms give differently shaped unit balls (a circle for the usual distance, a square with horizontal sides for $\max(|x|, |y|)$), and a first course in analysis or metric spaces shows that on $\mathbb{R}^2$ they all define the same notions of convergence and continuity. Bartle and Sherbert's *Introduction to Real Analysis* is a good place to meet these ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.